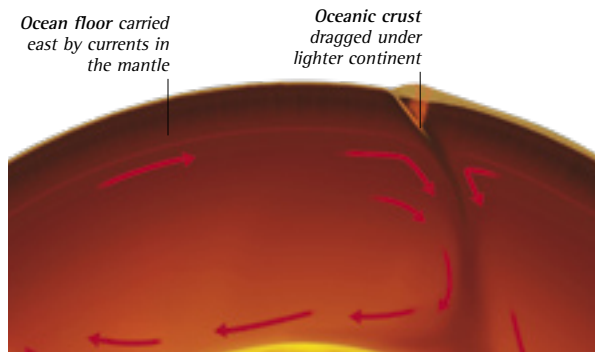
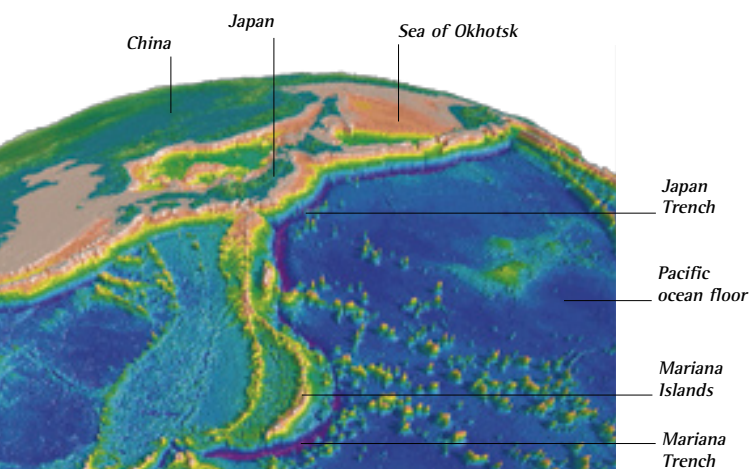


OCEAN TRENCHES

As new sections of ocean floor are formed at midocean ridges, some oceans like the Atlantic grow wider each year. But the planet is not getting any bigger. While some ocean floors expand, others shrink. Their edges sink back into the hot mantle along destructive plate margins known as subduction zones. The process creates deep ocean trenches and long lines of volcanoes, pushes up mountain ranges, and causes earthquakes and tsunamis.

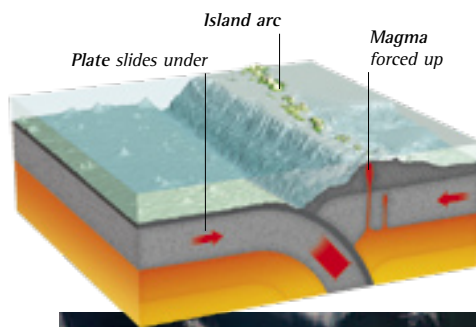


▲ DESTRUCTIVE MARGINS

Oceanic crust that is formed at a midocean ridge gradually spreads away from the ridge, driven by currents in the mantle below. This may simply make the ocean wider, but in the eastern Pacific, for example, oceanic crust moves east until it reaches the Peru–Chile Trench off South America. Here, it is dragged beneath the lighter crust of the continent and eventually destroyed.

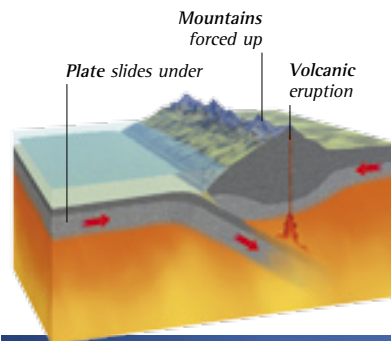
◀ SUBMARINE CHASMS

The subduction zones where the crust is destroyed are marked by deep ocean trenches, often associated with chains of volcanic islands. These trenches have been created by the ocean floor being dragged down into the hot heart of the planet by descending currents in the mantle. Although they are partly or even completely filled with sediments, some can be three times as deep as the nearby ocean floor. The lowest point of the Mariana Trench in the western Pacific lies nearly 7 miles (11 km) below the ocean surface, making it the deepest chasm on Earth.



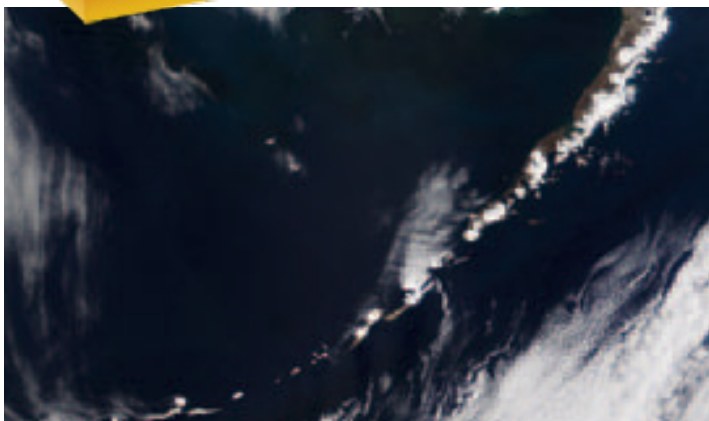
◀ OCEANIC BOUNDARY

As a subducted plate of oceanic crust plunges beneath another plate and into the hot mantle, it melts. The molten magma erupts through volcanoes in the overlying plate margin, and these often form long, curved chains of volcanic islands called island arcs.



◀ CONTINENTAL BOUNDARY

Where the ocean floor is being dragged beneath a continent, such as South America, the friction buckles the continental fringe into high mountains like the Andes. The volcanoes that would form island arcs at an oceanic boundary erupt through these mountains.



▲ ISLAND ARCS

Volcanic island arcs trace the plate margin in a curved chain like the Aleutian island arc in the north Pacific, seen here from space. Over time, the islands get bigger and join together, to form elongated islands like Java on the Sunda Arc.



▲ MOUNTAIN RANGES

The fold mountains thrown up along continental boundaries are initially high and rugged, like the Andes seen here. They are dotted with volcanoes, and the rocks often contain rare, valuable deposits such as copper, silver, and emeralds.